

Lecture Scribe L12

L12_S1_A.pdf

PDF

Example Lecture Scribe.pdf

PDF

Task Instructions: Using the same structure as the example provided above, generate a comprehensive lecture scribe based on the attached context for Lecture 12(L12_S1_A.pdf). This scribe is authored by Vansh Lilani (ID: AU2320146). It is intended to serve as a primary exam-oriented reference material for a reading-based/closed-notes exam.

Requirements:

Faithfulness & Detail: Include all definitions, notation, and assumptions exactly as presented. Every formula, theorem statement, and worked example from the lecture must be included in full.

Step-by-Step Explanations: For all proofs and derivations, provide the complete step-by-step logic shown in the slides. Do not summarize; ensure the explanation covers the transition between each mathematical step to ensure exam readiness.

Scope: Stick strictly to the provided PDF. Do not invent new content or add external "intuition" not present in the lecture materials.

Correctness & Tone: Maintain a formal academic tone. Prioritize mathematical precision and clarity in the layout of formulas.

Structure: Adhere strictly to the organizational pattern (headings, lists, and layout) demonstrated in the "Example Lecture Scribe" provided above.

Here is the lecture scribe written using the **same structure as the example lecture scribe** and based strictly on the uploaded lecture slides.

CSE400 — Fundamentals of Probability in Computing

Lecture L11-12: Transformation of Random Variables

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(Structure adapted from the provided example scribe.)

(Content derived from Lecture L12_S1_A slides.)

Example Lecture Scribe

L12_S1_A

1. Topic Title

Transformation of Random Variables and Functions of Two Random Variables

Topics covered:

- Transformation of random variables

- Function of two random variables
- Derived distribution example: $Z = X + Y$

(Outline slide — page 2)

2. Definitions and Notation

Let:

- X be a continuous random variable with known PDF $f_X(x)$.
- $F_X(x)$ be the CDF of X .
- A transformation is defined as:

$$Y = g(X)$$

If invertible:

$$x = g^{-1}(y)$$

Goal:

$$f_Y(y) \text{ and } F_Y(y)$$

For two random variables:

$$Z = g(X, Y)$$

Examples mentioned in lecture:

$$Z = X + Y, X - Y, \frac{X}{Y}, XY$$

(page 1 and page 3)

3. Assumptions / Conditions

Monotonic transformation

The function $g(x)$ must be:

- monotonically increasing OR
- monotonically decreasing

(Page 3 diagram explanation)

4. Main Results / Theorems

Transformation theorem (single RV)

Step-1: CDF transformation

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(g(X) \leq y) \end{aligned}$$

If g is increasing:

$$\begin{aligned} F_Y(y) &= P(X \leq g^{-1}(y)) \\ &= F_X(g^{-1}(y)) \end{aligned}$$

If g is decreasing:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

(pages 3-4)

Step-2: PDF transformation

Differentiate CDF:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Result:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$$

where

$$x = g^{-1}(y)$$

(page 3)

5. Proofs / Derivations

Function of Two Random Variables

Let

$$Z = X + Y$$

CDF definition:

$$\begin{aligned} F_Z(z) &= P(Z \leq z) \\ &= P(X + Y \leq z) \end{aligned}$$

(page 7)

Integral representation

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y) dx dy$$

(page 7)

Leibniz Rule (used in derivation)

If

$$G(x) = \int_{a(x)}^{b(x)} h(x, y) dy$$

then

$$\frac{d}{dx} G(x) = \frac{db(x)}{dx} h(x, b(x)) - \frac{da(x)}{dx} h(x, a(x)) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} h(x, y) dy$$

(page 8)

Differentiating CDF

Using Leibniz rule:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

After simplification:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(z - y, y) dy$$

Equivalent form:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x) dx$$

(page 9)

Independent case (Convolution)

If X and Y are independent:

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

Therefore,

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

This is called the **convolution integral**.

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6. Worked Examples

Example 1 — Transformation of RV

Given:

$$X \sim \text{Uniform}(-1, 1)$$

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find PDF of:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

(page 5)

Solution

Inverse transform:

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Derivative:

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

PDF formula:

$$\begin{aligned} f_Y(y) &= f_X(x) \left| \frac{dx}{dy} \right| \\ &= \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}} \\ f_Y(y) &= \frac{1}{\pi \sqrt{1-y^2}}, \quad -1 < y < 1 \end{aligned}$$

(page 5)

Example 2 — Sum of independent normals

Given:

$$X \sim N(0, 1), \quad Y \sim N(0, 1)$$

Independent.

(page 10)

Convolution

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-(z-x)^2/2} dx$$

(page 10)

After completing squares (shown on page 11):

$$Z \sim N(0, 2)$$

Example 3 — Exponential RVs

Let

$$X, Y \sim \text{Exponential}(\lambda)$$

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y > 0$$

(page 12)

Convolution

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$f_Z(z) = \lambda^2 z e^{-\lambda z}, \quad z > 0$$

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Example 4 — Difference of RVs

$$Z = X - Y$$

CDF:

$$F_Z(z) = P(X - Y \leq z)$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{z+y} f_{XY}(x, y) dx dy$$

Then

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

(page 13)

Additional transformations mentioned

- $Z = \frac{X}{Y}$
- $R = \sqrt{X^2 + Y^2}$

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If you'd like, I can:

- convert this into a **submission-ready PDF**, or
- format it exactly like your **previous lecture scribes**, or
- help you create a **1-page exam cheat-sheet** from this lecture.

Sources